SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : PRINCIPLES OF OPERATING SYSTEM

Practical No : 6

Aim :
CPU Scheduling Algorithms (Part 2) – Round Robin
Implement Round Robin scheduling with configurable time quantum.
Compare with FCFS: fairness, turnaround, response time.
Track context switches and improve queue management.

CODE 6A :
```
from collections import deque

print("Om Wala S119")

processes = [
    ["P1", 0, 5],
    ["P2", 4, 2],
    ["P3", 5, 4]
]

time_quantum = 2

def round_robin(processes, tq):

    remaining = {}
    arrival = {}
    burst = {}
    completion = {}

    for p in processes:
        remaining[p[0]] = p[2]
        arrival[p[0]] = p[1]
        burst[p[0]] = p[2]

    queue = deque()
    gantt = []

    time = 0
    i = 0
    n = len(processes)

    processes = sorted(processes, key=lambda x: x[1])


    while i < n and processes[i][1] <= time:
        queue.append(processes[i][0])
        i += 1

    while queue:
```

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```
        pid = queue.popleft()

        start = time
        run_time = min(tq, remaining[pid])
        time += run_time

        gantt.append((pid, start, time))

        remaining[pid] -= run_time


        while i < n and processes[i][1] <= time:
            queue.append(processes[i][0])
            i += 1

        if remaining[pid] > 0:
            queue.append(pid)
        else:
            completion[pid] = time

    turnaround = {}
    waiting = {}

    for pid in arrival:
        turnaround[pid] = completion[pid] - arrival[pid]
        waiting[pid] = turnaround[pid] - burst[pid]

    avg_tat = sum(turnaround.values()) / n
    avg_wt = sum(waiting.values()) / n

    return gantt, turnaround, waiting, avg_tat, avg_wt



def fcfs(processes):

    processes = sorted(processes, key=lambda x: x[1])

    time = 0
    gantt = []
    turnaround = {}
    waiting = {}

    for p in processes:

        pid = p[0]
        arrival = p[1]
        burst = p[2]
```

```
        if time < arrival:
            time = arrival

        start = time
        time += burst

        gantt.append((pid, start, time))

        turnaround[pid] = time - arrival
        waiting[pid] = turnaround[pid] - burst

    n = len(processes)

    avg_tat = sum(turnaround.values()) / n
    avg_wt = sum(waiting.values()) / n

    return gantt, turnaround, waiting, avg_tat, avg_wt



def display(title, gantt, turnaround, waiting, avg_tat, avg_wt):

    print("\n" + "=" * 50)
    print(title)
    print("=" * 50)

    print("\nGantt Chart:")
    for pid, start, end in gantt:
        print(f"| {pid} {start}-{end} ", end="")
    print("|")

    print("\nTurnaround Time:")
    for pid in turnaround:
        print(f"{pid} = {turnaround[pid]} ms")

    print("\nWaiting Time:")
    for pid in waiting:
        print(f"{pid} = {waiting[pid]} ms")

    print(f"\nAverage Turnaround Time = {avg_tat:.2f} ms")
    print(f"Average Waiting Time = {avg_wt:.2f} ms")


rr_gantt, rr_tat, rr_wt, rr_avg_tat, rr_avg_wt = round_robin(
    processes, time_quantum
)
```

```
display(
    "ROUND ROBIN (Time Quantum = 2 ms)",
    rr_gantt,
    rr_tat,
    rr_wt,
    rr_avg_tat,
    rr_avg_wt
)

fcfs_gantt, fcfs_tat, fcfs_wt, fcfs_avg_tat, fcfs_avg_wt = fcfs(
    processes
)

display(
    "FCFS SCHEDULING",
    fcfs_gantt,
    fcfs_tat,
    fcfs_wt,
    fcfs_avg_tat,
    fcfs_avg_wt
)
```

OUTPUT :

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IDLE Shell 3.14.6

File Edit Shell Debug Options Window Help

```
Om Wala S119

==================================================
ROUND ROBIN (Time Quantum = 2 ms)
==================================================

Gantt Chart:
| P1 0-2 | P1 2-4 | P2 4-6 | P1 6-7 | P3 7-9 | P3 9-11 |

Turnaround Time:
P1 = 7 ms
P2 = 2 ms
P3 = 6 ms

Waiting Time:
P1 = 2 ms
P2 = 0 ms
P3 = 2 ms

Average Turnaround Time = 5.00 ms
Average Waiting Time = 1.33 ms

==================================================
FCFS SCHEDULING
==================================================

Gantt Chart:
| P1 0-5 | P2 5-7 | P3 7-11 |

Turnaround Time:
P1 = 5 ms
P2 = 3 ms
P3 = 6 ms

Waiting Time:
P1 = 0 ms
P2 = 1 ms
```

```
Average Turnaround Time = 4.67 ms
Average Waiting Time = 1.00 ms
```

CODE 6B :

```
from collections import deque

print("Om Wala S119")

processes = [
    ["P1", 0, 5],
    ["P2", 1, 3],
    ["P3", 2, 6]
```

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```
]

time_quantum = 2

def round_robin(processes, tq):

    processes = sorted(processes, key=lambda x: x[1])

    n = len(processes)
    remaining = {}
    arrival = {}
    burst = {}
    completion = {}

    for p in processes:
        remaining[p[0]] = p[2]
        arrival[p[0]] = p[1]
        burst[p[0]] = p[2]

    queue = deque()
    gantt = []

    time = 0
    i = 0


    while i < n and processes[i][1] <= time:
        queue.append(processes[i][0])
        i += 1

    while queue:

        pid = queue.popleft()

        start = time


        run_time = min(tq, remaining[pid])

        time += run_time
        remaining[pid] -= run_time

        gantt.append((pid, start, time))


        while i < n and processes[i][1] <= time:
            queue.append(processes[i][0])
            i += 1
```

```
        if remaining[pid] > 0:
            queue.append(pid)
        else:
            completion[pid] = time


    turnaround = {}
    waiting = {}

    for p in processes:
        pid = p[0]

        turnaround[pid] = completion[pid] - arrival[pid]
        waiting[pid] = turnaround[pid] - burst[pid]

    avg_tat = sum(turnaround.values()) / n
    avg_wt = sum(waiting.values()) / n

    return gantt, turnaround, waiting, avg_tat, avg_wt


gantt, turnaround, waiting, avg_tat, avg_wt = round_robin(
    processes,
    time_quantum
)


print("\nRound Robin Scheduling")
print("Time Quantum =", time_quantum, "ms")

print("\nGantt Chart:")

for pid, start, end in gantt:
    print(f"| {pid} {start}-{end} ", end="")

print("|")


print("\nTurnaround Time:")

for pid in turnaround:
    print(f"{pid} = {turnaround[pid]} ms")


print("\nWaiting Time:")
```

```python
for pid in waiting:
    print(f"{pid} = {waiting[pid]} ms")


print("\nAverage Turnaround Time =", round(avg_tat, 2), "ms")
print("Average Waiting Time    =", round(avg_wt, 2), "ms")
```

OUTPUT :

```
Om Wala S119

Round Robin Scheduling
Time Quantum = 2 ms

Gantt Chart:
| P1 0-2 | P2 2-4 | P3 4-6 | P1 6-8 | P2 8-9 | P3 9-11 | P1 11-12 | P3 12-14 |

Turnaround Time:
P1 = 12 ms
P2 = 8 ms
P3 = 12 ms

Waiting Time:
P1 = 7 ms
P2 = 5 ms
P3 = 6 ms

Average Turnaround Time = 10.67 ms
Average Waiting Time    = 6.0 ms
>>>
```